

Pollution and its Impact on Predator Cue Detection

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Abstract

Chemical cues excreted by predators are essential for prey evasion and survival, yet anthropogenic pollutants may disrupt this process¹. The shore crab (*Hemigrapsus oregonensis* and *Hemigrapsus nudus*) coexist with the invasive predator the European green crab (*Carcinus maenas*) in the Puget Sound where contaminated runoff is common². While observed in other organisms, it is unknown whether pollutants impair the crab's ability to detect predator cues from, specifically from the growing threat of the invasive species. The shore crabs were exposed to four treatments: a control, only predator cues, only pollution (low and high levels), and pollution combined with predator cues (low and high levels). Metabolic rate, measured with resazurin fluorescence³, and number of antennae flicks per minute were measured as a proxy to their stress response when exposed to a predator. Under high levels of pollution, the metabolic rate in the pollution and predator cues treatment was significantly lower than the treatment with predator cues ($p < 0.05$). However, the antennae flicking under high pollution and predator cues was significantly higher than the high pollution only group ($p < 0.05$). No other comparisons were significant. These contrasting results suggest that pollutants impair the metabolic response to predator cues while still exhibiting a behavioral chemosensing response. These findings indicate that Puget Sound pollutants may still disrupt predator detecting in native shore crabs, possibly increasing their vulnerability to an invasive predator.

Introduction

The Puget Sound is home to several major species of crabs both big and small, each playing a major role in the balance of the ecosystem. These species include the hairy and purple shore crab (*Hemigrapsus oregonensis* and *Hemigrapsus nudus*) that control plants and small invertebrate populations as well as act as a critical food source for larger native crabs, fish, and birds. However, with the increase in the presence of the invasive European green crab (*Carcinus maenas*) since its introduction in 1998, they have been known to prey on many of the native crabs dramatically harming their population.

To avoid predation events, shore crabs rely heavily on their ability to sense chemical cues released by predators in the water, known as chemosensing. Antennae flicking is a direct behavioral measure of chemosensing as crabs will increase their flicking rate when they detect predator cues, allowing them to assess the possibility of a threat and respond accordingly¹. Metabolic rate also serves as an indirect indicator of stress, as exposure to predator cues often elevates the metabolic processes within an organism in preparation for a fight-or-flight type response⁴.

Unfortunately, the Puget Sound is consistently subjected to chronic pollution from urban and industrial runoff which includes heavy metals (e.g., copper, zinc, aluminum), road and yard runoff, and other major contaminants². These pollutants are known to impair chemosensory functions in a variety of aquatic organisms from fish to crustaceans⁵. For example, copper exposure reduces olfactory sensitivity in salmon and disrupts antennular flicking in lobsters¹. However, it remains unknown whether environmentally relevant levels of Puget Sound

pollutants interfere with the native shore crab's ability to detect and respond to chemical cues from the invasive predatory European green crab.

In this pilot study, the effects of pollutants were tested on two physiological proxies of chemosensing in shore crabs. Metabolic rate via resazurin fluorescence and the antennae flicking frequency. Crabs were posted to four treatments with two different levels of pollution (low and high): clean seawater (control), predator cues only, pollution only, and pollution combined with predator cues. It is predicted that pollutants would reduce both metabolic and behavioral responses to predator cues with stronger effects at higher pollutant concentrations. Specifically, it is hypothesized that (1) predator cues alone would increase both metabolic rate and flicking relative to the control, (2) pollution alone would have no effect or a suppressive effect, and (3) the combination of pollution and predator cues would result in lower responses than predator cues alone indicating chemosensory disruption.

Methods

Each of the treatment groups contained three crabs with two hairy shore crabs and one purple shore crab (12 crabs used in total). The groups of three were placed in small jars with a lid allowing water to cycle in while completely covered to keep the environment controlled. Two jars were placed in a bucket for the clean seawater groups, and another bucket for the polluted groups. The high polluted water contained 500µg/L of herbicides, 500µg/L of motor oil, 6.849g, and a 3in x 3in of aluminum foil with one side sprayed in two coats of copper based anti-fouling spray. This pollution treatment was used for one week and then switched to a lower dose with 150µg/L of herbicides and 150µg/L of motor oil for the second week. Crabs were then placed in each solution for a week before testing on a water table with a controlled temperature and constant oxygen.

After letting rest in the solution for a week, the jars were removed from the buckets and the antennae flicking rate was measured for both non-predator cue solution jars. A phone was used to record for a minute and then the counting was done afterwards by looking back at the footage. The data was then recorded in a datasheet. Once the antennae flicking was recorded for the two groups, resazurin was loaded into a 6-well plate as well as two plastic cups for the blanks. The crabs for each group were placed in the wells and let sit for an hour total. Using a micropipette, 2µg/L of the resazurin were loaded into the corresponding well in a 96-well plate for later fluorescence measurements. Resazurin measurements were taken at the initial placement in the well (T0), after 30 minutes (T1), and after a total hour (T2).

Afterwards, a single European green crab was placed in each of the buckets for ~30 minutes to allow for predator cue detection in the two jars. After 30 minutes, the jars were removed and the antennae and resazurin data were repeated. This same procedure was repeated for the low pollution group.

Once fluorescence was measured, the data was standardized using the blanks and the size of each of the crabs used for the experiment. Using the standardized values and the antennae flicking per minute, an ANOVA and Tukey test was run to analyze the significance in the difference between the groups.

Results

Metabolic rate, measured as resazurin fluorescence and standardized by the individual crab's size, varied across treatment combinations (Figure 1). The figure shows that predator cues increased the overall metabolic rate in the no and low pollution groups while decreasing in the high pollution group. Using an ANOVA test revealed a significant overall effect of the treatment ($p = 0.01$). Using a Tukey test to analyze the significance of each comparison revealed that the predator cue only treatment elicited a significantly higher metabolic rate than the high pollution and predator cues ($p = 0.026$). No other pairwise comparison reached statistical significance. While visually distinct, the control group did not differ from any treatment.

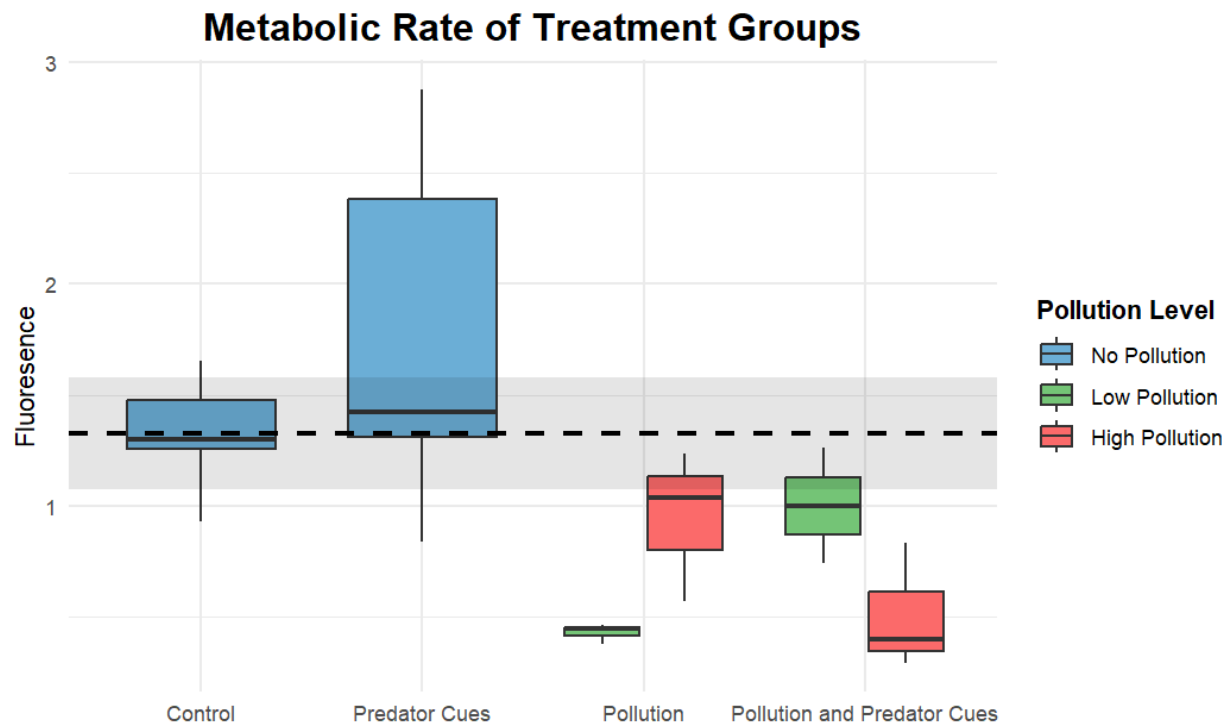


Figure 1 The final fluorescence (T2) in each of the four treatment groups after standardization. Groups are colored based on pollutant exposure level. Dashed line and highlighted region represents the mean control value and the standard deviation for more accurate comparison.

Antennae flicking per minute showed varying effects across the different treatments (Figure 2). The control and predator cues were similar with the only major differences in antennae flicking being seen in the high pollution with no predator cues, and the low pollution with predator cues. Both are notably lower than any corresponding group. The overall ANOVA was significant ($p = 1.21e-5$). The Tukey test revealed several key significant comparisons; however, the major notable difference is that antennae flicking in the high pollution with predator cues is significantly higher than the high pollution only group ($p = 5.4e-5$). The results also showed that higher pollution levels significantly suppressed antennae flicking ($p = 1.0e-4$). No other comparisons were statistically significant.

Mean Antennae Flicking by Treatment and Pollution Level

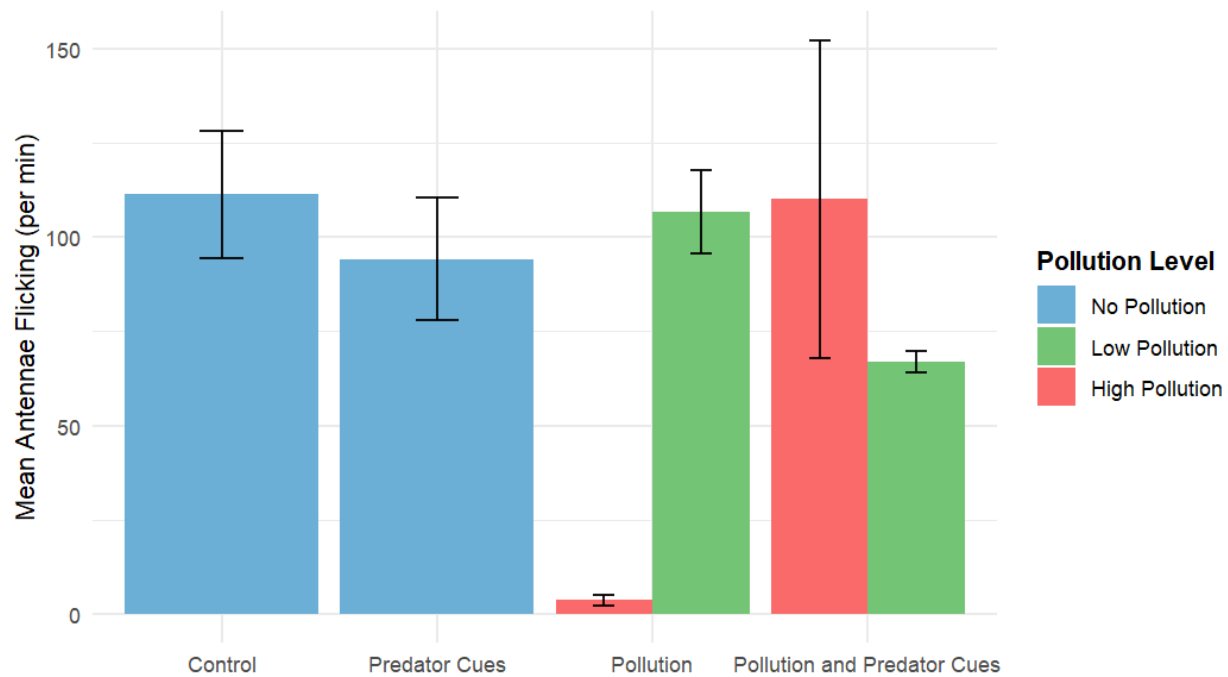


Figure 2 The mean antennae flicking per minute in each of the four treatment groups with the standard deviation represented in the error bars. Groups are colored based on pollutant exposure level.

Discussion

This study investigated if pollutants commonly found in Puget Sound disrupt chemosensing in native shore crabs (*Hemigrapsus oregonensis* and *Hemigrapsus nudus*) in response to cues from the invasive European green crab. Two proxies were used to assess the impact: metabolic rate through resazurin fluorescence as an indirect measure of physiological stress, and antennae flicking as a direct behavioral measure of chemosensory sampling. The results provide partial support for the alternative hypothesis: pollutants altered the crab's response to predator cues, but the effects varied between the two proxies.

Predator cues alone significantly elevated metabolic rate compared to both low pollution and high pollution with predator cue treatments. This suggests that when pollution is absent, crabs detected the predator cues from the European green crabs and physiologically responded with an increased stress response¹. However, pollution alone did not significantly alter metabolic rate relative to the control which indicates that the pollutants themselves did not impact the metabolic rate. Instead, the disruption appears to originate from the chemosensory to metabolic cascade which aligns with previous findings that heavy metal can impair olfactory ability⁶.

The antennae flicking results showed varying effects across treatments. The results indicate that pollution, especially at high concentrations, generally suppresses antennae flicking behavior which is a direct measure of chemosensory sampling. This aligns with previous findings that heavy metals can damage olfactory sensory abilities⁶. Surprisingly, when high pollution was combined with predator cues, flicking was significantly higher than in high

pollution alone which suggests that the predator cues partially override the suppressive effect of pollution. This fact may indicate that there is an interactive effect between pollution and predator cues that warrants further investigation.

This divergence between the metabolic effect and behavioral response was not hypothesized but noteworthy. A possible explanation is that the metabolic rate reflects a longer, whole body stress response whereas the antennae flicking is a rapid and reflexive behavior that can be triggered even when physiological responses are compromised.

The European green crab is a voracious predator that has already decimated native shellfish and crab populations along the US West Coast. The findings in this study suggest that high levels of pollution could possibly impair the ability of shore crabs to detect and respond to this invasive predator. The loss of the predator-induced metabolic increase under high pollution may leave crabs energetically drained leading to more successful predation events by the green crab. However, the partial preservation of flicking behavior under the same conditions may still allow some threat assessment in the shore crabs. It remains to be tested whether these effects translate to reduced survival in wild populations.

This study only employed a handful of pollutants and temporary exposure, and the pollutant levels were greatly exaggerated. Chronic, low exposure levels may produce different effects on the shore crabs compared to the acute high exposure levels seen in this experiment. Additionally, only proxies for physiological stress and sensory were measured instead of direct stress hormones which could possibly lead to more insight into the response from the shore crabs. Future studies should analyze and better replicate the conditions experienced by wild populations as well as a more detailed analysis of the physiological response.

Despite the limitations, Puget Sound pollutants disrupt chemosensing in native shore crabs but in a complex and slightly contradictory way. High pollution alone suppresses antennae flicking, yet the presence of predator cues restores this behavior while simultaneously abolishing the normal metabolic increase. These findings indicate that pollution can disrupt behavioral and physiological components of the shore crab's predator response, leaving them potentially increasingly vulnerability to invasive predators. As the European green crab continues to spread, further understanding of how environmental stressors influence predator-prey interactions is critical for conservation and environmental management.

References

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